Classic Modes of Data Collection

Fundamentals of Data Collection I
October 28, 2024
Notes by Frederick Conrad

Outline

- Modes of Data Collection
- Computer Assisted Survey Information Collection
- Effects of Modes on Survey Error
  - Sampling Frames & Coverage
  - Nonresponse
  - Measurement
- Mode Effects

Survey Mode: Combination of Medium and Agent

- Combination of *medium* and *agent*
- Medium (channel of communication):
  - live voice (in person or via phone or live video)
  - text (on a screen or on paper)
  - prerecorded or synthesized speech
- Agent (who administers the questionnaire):
  - live interviewer
  - respondent (self-administers questionnaire)
  - prerecorded or animated interviewer, chatbot
- Other agents and media certainly possible

Some Possible Modes

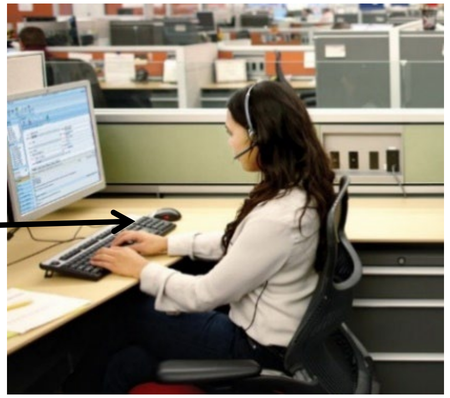
- Some possible modes:
  - Interviewer + (live) voice: in person, i.e., face-to-face, interview
  - Interviewer + (live) voice: telephone interview
  - Respondent + paper: mail questionnaire
  - Respondent + onscreen text: web survey
  - Respondent + (prerecorded) voice: "robocall," Interactive Voice Response (IVR)
  - Prerecorded interviewer + onscreen video: virtual interviewer (?)
- If *R* is agent, mode is considered *self-administered*
- If *Iwer* is agent, mode is considered *Iwer-administered*
- If animated *Iwer* asks each question when *R* presses a button, is this *Iwer*- or *self*-administered?

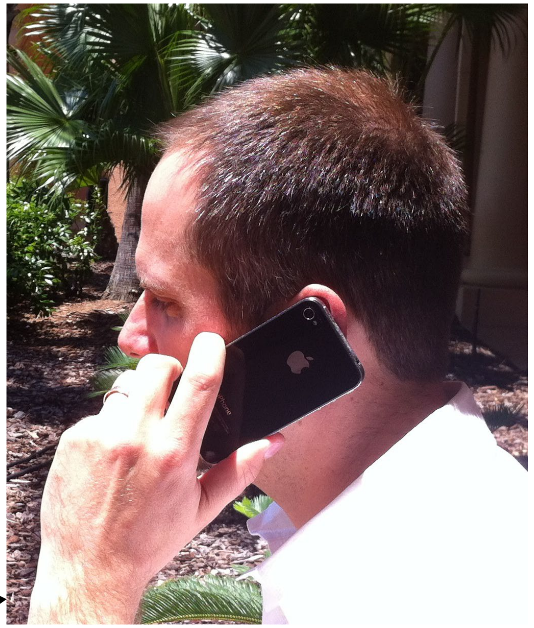
Now Many Modes of Data Collection

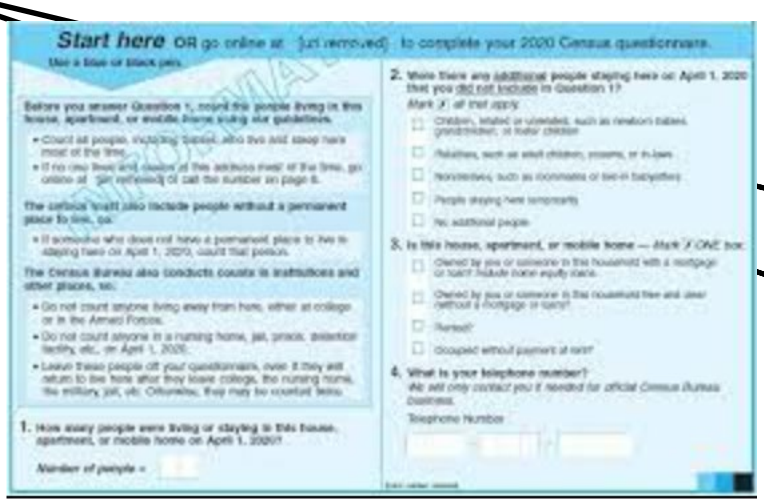
- Until recently, there were only a few plausible modes:
  - In-person (or face-to-face [FTF] or personal) interviews
  - Mail (or postal) surveys
  - (landline) Telephone surveys
- When computers are part of the mode, convention used to be to denote this in the name of the mode by including "CA"
  - e.g., CATI = Computer Assisted telephone Interviewing
- In last few years, explosion of methods, most notably online ("web")
- Variants of existing modes:
  - Mobile phone interviews (location not tied to phone number as with landlines, device is associated with an individual instead of a household)
  - Mobile web: self-administered, text-based questionnaires on smart phone in browser or app

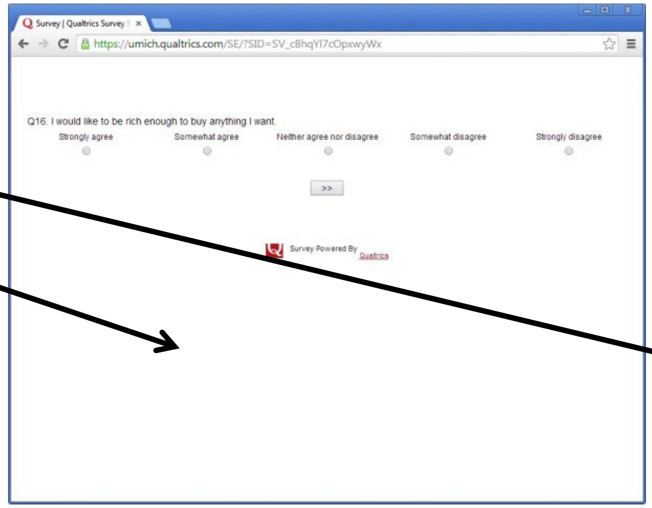
Modes of Data Collection

- Major modes:
  - CATI*
  - CAPI**
  - Web
  - Mobile Web
  - Mobile Phone
  - Mail









*Computer Assisted Telephone Interview
**Computer Assisted Personal Interview

Some Emerging Modes

- Text Message (SMS)
  - Self-administered (automated) or interviewer-administered
  - High coverage rates – mobile phones support texting even if not internet-enabled (smartphone)
- Live Video
  - e.g., Zoom, MS Teams, Google Meet, FaceTime, etc.
  - How similar is live video interviewing to in-person (FTF) interviewing?
- Virtual interviewers
  - Video or computer-animated interviewing agents
  - "autonomous" vs. recorded agents
  - Embodied vs. language-only (Chatbots)
- *Note:* More about emerging modes in week 10

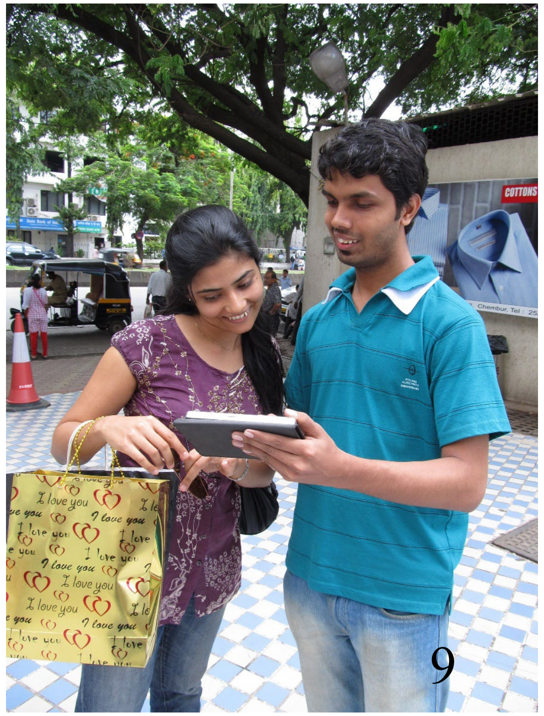
Researchers' Considerations when Choosing a Mode

- Sometimes the choice is dictated by the topic
  - *e.g., what mode is needed for a literacy survey?*
- Sometimes the choice is driven by the sampling frame
  - *e.g., how do you survey voters on election day?*
- Sometimes the decision is driven by cost, timeliness, resources, etc.
  - *e.g., if there are not sufficient resources to hire Iwers, what self-administered mode is best?*
- Within these broad options many decisions required
  - *e.g., should in-person Iwer present show card to respondents?*

Personal (FTF) Interviewing Developments

- Original interview mode:
  - *Iwer* read questions from paper q'aire aloud and wrote responses on the q'aire
- Now, in-person (FTF) *Iwer* usually reads *Q* from screen and enters responses by clicking or typing: Computer Assisted Personal Interviewing (CAPI)
- Widespread use starting in 1990s, once portable computers – laptops – really became portable



Telephone Interviewing Developments

- Developed by U.S. market research organizations and academic survey centers in late 1960s-1970s; governments followed in 1980s
- Today, "telephone survey" probably means *Iwer* reads questions from screen to *R* via phone: Computer Assisted Telephone Interview or CATI
- CATI used worldwide; centralized calling facilitates monitoring interviews

- Primarily centralized but can also be decentralized
  - *Iwer* can phone *R*, driving interview with laptop provided for in-person data collection

In-Person (FTF) Telephone

CAPI vs. CATI

- Hardware: typically provided by organization
- Direct supervision/monitoring: most feasible in centralized facility
- Physical environment and connectivity: standing on doorstep or in crowded space vs. call center
- Presence of computer during interview (effect on *I*-*R* interaction): less of an issue when interview conducted remotely

Computer Assisted Self Interviewing (CASI)

- Part of an interviewer-administered survey
- *Iwer* asks non-sensitive questions
- *R* self-administers questions on sensitive topics on same computer
- Flavors of CASI
  - CASI: text-based or at least visual, sometimes called “text-CASI” for clarity
  - Audio-CASI: questions audio-recorded, *R* listens with headphones/ear buds and enters answers with keypad/pointing device
  - Telephone audio-CASI (T-ACASI) also called recruit-and-switch or outbound IVR*: live *Iwer* asks non-sensitive *Q*s over phone and switches *R* to automated administration for sensitive *Q*s

*Interactive Voice Response

Interactive Voice Response (IVR)

Outbound (recruit-and-switch)*

- *Iwer* initiates call
- Persuades sample member who answers phone to participate or provide HH roster, collects basic screening info, randomly selects *R* from HH
- Asks non-sensitive *Q*s
- Switches *R* to IVR for remaining, usually sensitive, *Q*s

*Also referred to as T-ACASI

Inbound

- *R* initiates calls
- No *I'wer* involvement, i.e., entirely self-administered
- Example: CES

- Touchtone IVR vs. Speech IVR

Effects of Mode on Survey Error

- No single “best mode” for all situations
- Decisions on data collection methods involve trade-offs on a variety of design dimensions
  - Sampling frame (coverage), sampling methods
  - Contact and recruitment methods (nonresponse)
  - Subject matter (sensitive or not?, measurement)
  - *Iwer*-administration generally increases response rates but is usually substantially more expensive than self-administered modes (especially web)

Mode and Sampling Frame

- A frame is a list of elements in target population
- Main types of frames:
  - List frame
  - Area frame
  - RDD frame
  - Intercept or transaction frame
- Limitation generalizing to target population due to
  - Coverage of frame given the mode
  - Ability of sample to use the mode
- Richer frame permits more modes of contact
  - e.g., street address permits advance letters, including prepaid incentives, to be mailed to sample member

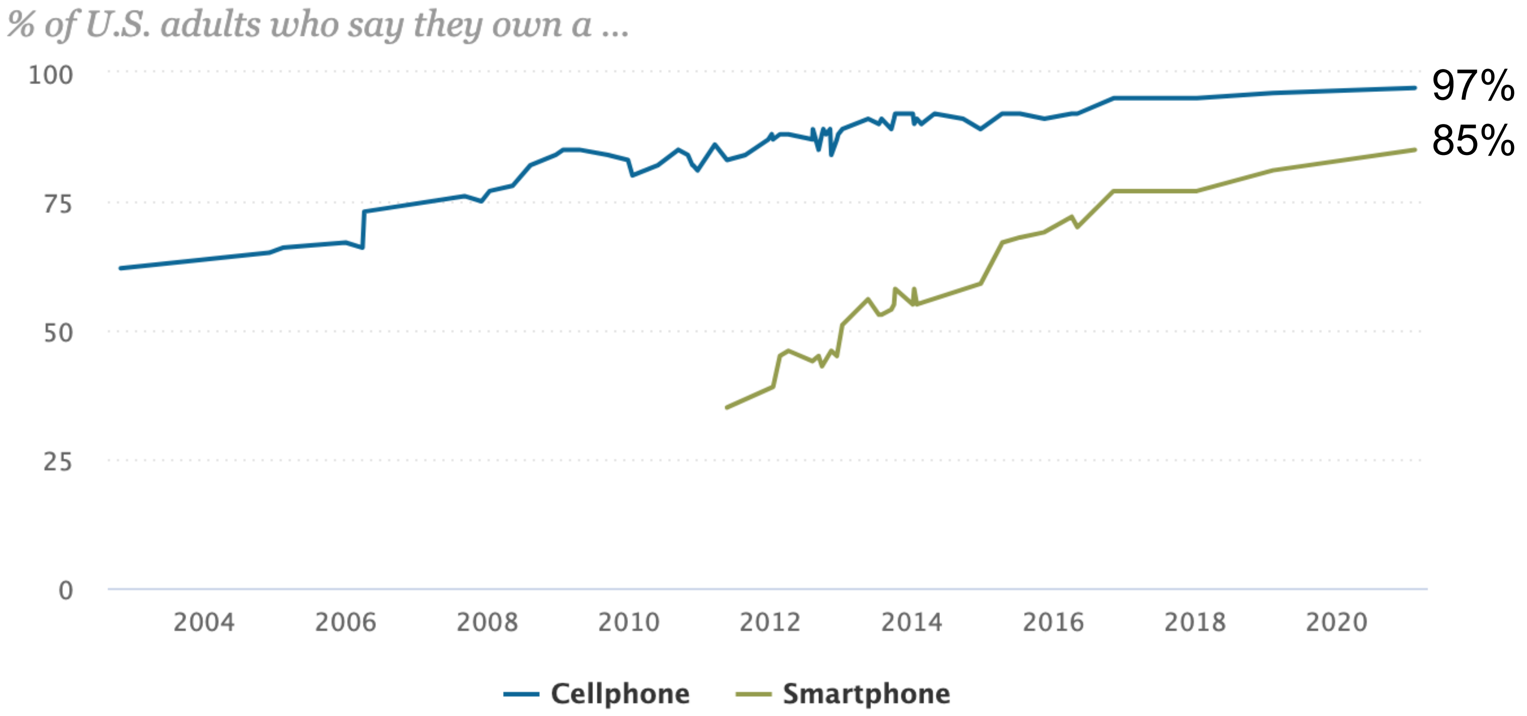
Sampling Frames for Mail and In-Person Data Collection

- Street addresses needed for both
- When no sampling frame of persons or *HH*s, solutions differ by country
  - in US, address based sampling (ABS) selects units from Delivery Sequence File (DSF) maintained by US Postal Service; augmented and sold by 3rd parties
  - In UK, Postcode Address File maintained by Royal Mail is list of all addresses in country, continuously updated
  - In many countries, population registers maintained by government
    - up-to-date list of all living residents, including addresses and demographics for each
- When address frame not available or practical, area frame is “listed”
  - Trained staff (often interviewers) walk/drive each sampled block and record (list) eligible addresses
  - Expensive, time-consuming, and prone to error

Sampling Frames for Telephone Surveys

- List frames (e.g., telephone directories)
- RDD generates sample of telephone numbers from all possible numbers
  - Very inefficient: ~0.4% of all possible numbers are working
  - Restricting to active prefixes increases rate to about ~16%
  - List-assisted approaches further improve efficiency
- Telephone frames now typically include designated cell-phone prefixes
  - Dual-frames
  - Changes sample unit from HH to individual
  - Federal Communications Commission ruled in July 2015 that autodialing mobile phone numbers requires prior consent (TCPA)

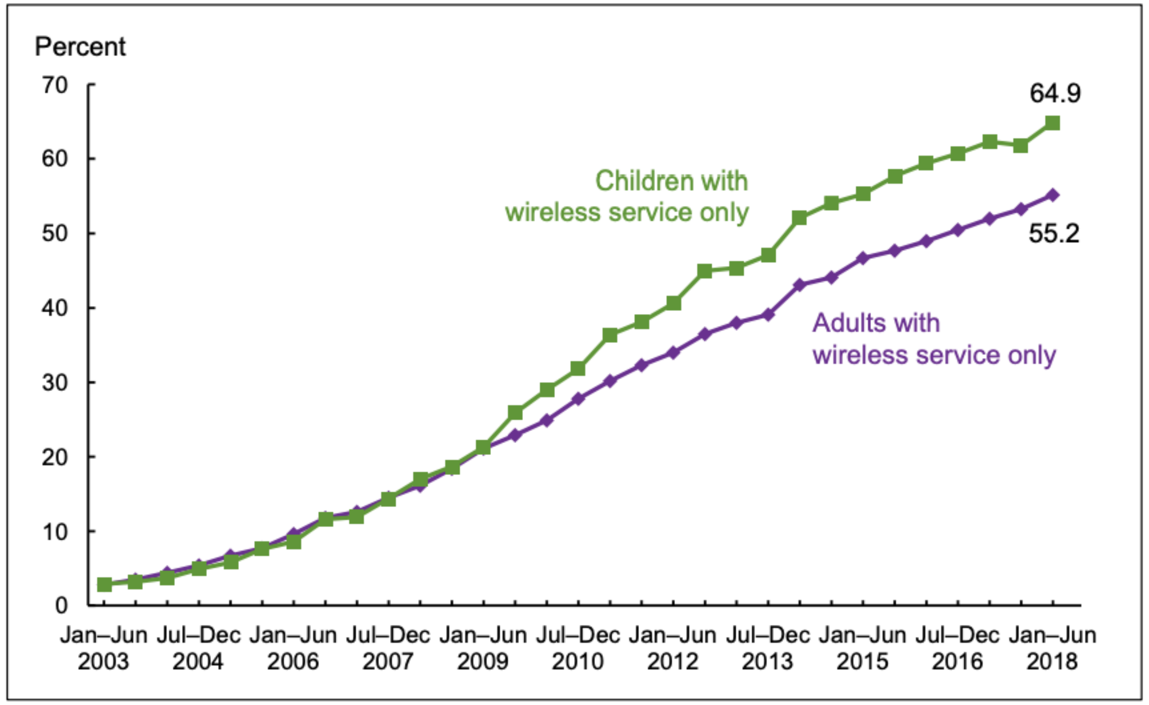
Mobile Phones and Coverage



Note: Respondents who did not give an answer are not shown.

Source: Surveys of U.S. adults conducted 2002-2021.

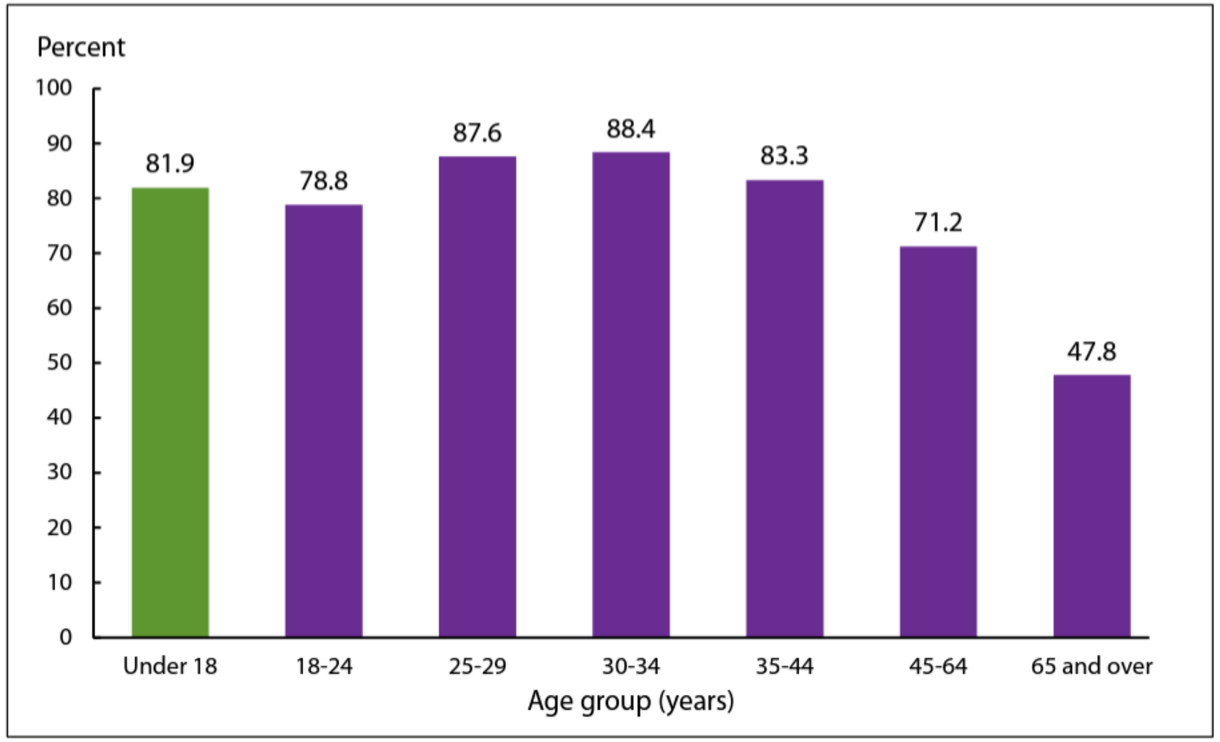
Figure. Percentages of adults and children living in households with only wireless telephone service: United States, 2003–2018



NOTE: Adults are aged 18 and over; children are under age 18.
DATA SOURCE: NCHS, National Health Interview Survey.

Mobile Phone Coverage Bias: Age and Income

Figure. Percentages of wireless-only adults and of children living in households with only wireless telephone service, by age group: United States, July–December 2022



NOTES: Wireless-only adults are adults who live in households with only wireless telephone service and have their own wireless telephone.
SOURCE: National Center for Health Statistics, National Health Interview Survey.

Adults with income below Federal poverty line more likely (74.4%) than those near poverty (67.7%) who are more likely than higher income adults (64.3%) to be wireless-only

Source: Blumberg SJ, Luke JV. Wireless substitution: Early release of estimates from the National Health Interview Survey, July-December 2022. National Center for Health Statistics. August 2021.

Mobile Phone Coverage Bias

- Blumberg & Luke (2015): Examples of bias among adults in NHIS Jul-Dec 2014

Prevalence Rates for Selected Health Behaviors/Status

	Landline *HHs*	Cell-only *HHs*	Phoneless *HHs*
≥1 heavy-drinking day in past year	19.6	30.3	25.4
Current smoker	12.7	20.2	21.3
Influenza vaccine during last year	48.5	34.5	34.3
Ever tested for HIV	31.5	44.2	38.3
Has usual place for medical care	90.8	80.2	79.7
Currently uninsured	10.9	18.7	19.7

Effect of Mode on Nonresponse

- Modes typically vary in response rates
  - can be -- but is not necessarily-- related to *nonresponse error*
- Modes vary in sources and types of nonresponse
  - Refusal vs. non-contact
  - Access restrictions, noncontact
  - Capacity/ability to do survey
- In mail and telephone surveys, can be hard to distinguish nonresponse (in particular non-contact) from coverage problem
  - e.g., no response to mailed questionnaire could be nonresponse or address could be ineligible (e.g., commercial, unoccupied), i.e., problem with frame
- As cell phone use has ↑, refusals have ↓ (Dutwin and Lavrakas, 2016); attributed to mobile users not answering calls (non-contact)

Nonresponse in Telephone Surveys

- Prevalence of mobile phones = prevalence of voice mail
- In Xu et al. (1993), *Iwer* called (landline) phones; if no answer:
  - either left up to three messages on answering machines or none (control)
  - Three messages led to 46% response rate vs. 33% if no message left
- Invitations more likely to be intercepted by answering machines (pre-voice mail) at certain times: especially Saturday night (Piazza, 1993)
  - Those with answering machines more difficult to reach, but once contacted, just as likely to participate
- How might these findings transfer to smartphone users ?

Nonresponse in IVR

- Breakoffs during outbound IVR common, especially at switch from interviewer to automated system
- Gribble et al. (2000)
  - Break off rate of 24% for T-ACASI, compared with 2% for CATI (i.e., where *Iwer* conducts entire interview, i.e., no switch to computer-administration)
- Tourangeau et al. (2002) report break-off rates as high as 31% in review of IVR studies
- Couper et al. (2003)
  - 23.6% overall break off rate in IVR
  - 10.5% broke off during transfer
  - Probably exacerbated by 7-second delay in transfer

Nonresponse in Mail Surveys

- Dillman (2000): *Total Design Method* (later *Tailored Design Method*) developed to increase response rates (RR) in mail surveys
  - "Respondent friendly" questionnaire
  - Up to five contacts (at least one "special," e.g., delivered via FedEx instead of USPS)
  - Return envelope including real stamp
  - Personalized correspondence
  - Token cash incentive sent with questionnaire
- Widely followed practice; has been adapted to web surveys
- Seems to work but "bundled" practices make it hard to know how each practice contributes to increased RR

Measurement Error: Interviewers' Influence

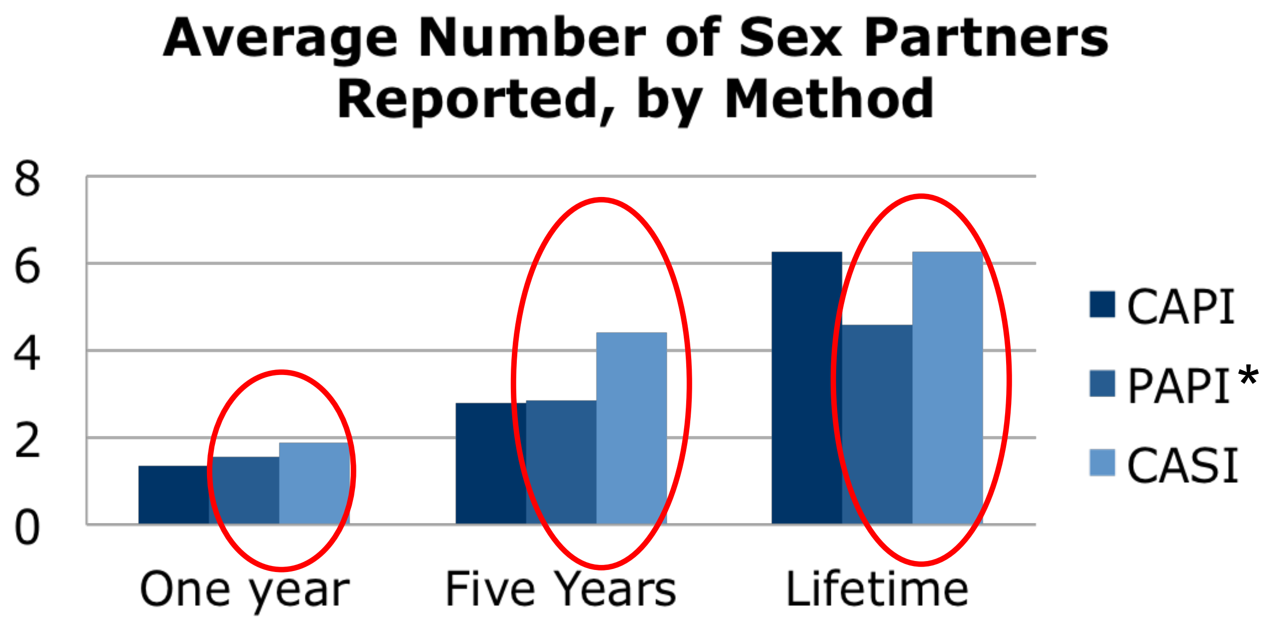
- *Iwers* not only achieve higher response rates, but they can also produce higher quality data
  - Motivate *R*s to adequately answer questions
  - Probe inadequate responses
  - Provide feedback, clarify questions and difficult terminology
- But they
  - can also increase socially desirable responses to *Q*s on sensitive topics
  - contribute variance due to how they conduct the interview vs. other *Iwer*s
  - are expensive compared to self-administered modes

Effect of CASI and Audio-CASI on Responses

- Less socially desirable responding for sensitive items in self-administered (CASI, ACASI) than *Iwer*-administered modes
  - observed whether answers to self-administered Qs recorded on paper or entered into computer/tablet
  - Generally assumed that more socially desirable responses = less candor
- ACASI, <u>compared to CASI</u>, presumed to
  - reduce literacy problems
  - increase privacy and thus candid responding
  - Little evidence that ACASI increases candid responding compared to CASI
  - Couper et al. (2009) find no increase in disclosure with ACASI vs. CASI but audio was not frequently used – so possible does provide more privacy

Effects of Computerization, Over and Above Self-Administration

- More sex partners disclosed (in sample of young women) in CASI than Paper and Pencil questionnaire (Tourangeau & Smith, 1998)



Source: Tourangeau & Smith, 1998

*PAPI= Paper and Pencil Interview

- Meta-analysis of studies comparing self-administration of sensitive questions by <u>computer</u> and <u>paper</u> finds that computerization resulted in 1.5 times higher prevalence of sensitive behaviors than P&P (Gnambs & Kaspar, 2014)
  - for highly sensitive issues, mode effect even larger

Summary: Main Survey Modes

- Mode is one of highest impact decisions researchers make when implementing research design
  - Requires considerations of all sources of error and costs
  - Decision should be based on both theory and empirical evidence
- Face-to-face, telephone, and self-administered surveys now largely computer-based (mail still common for some populations and for invites)
- Coverage error is function of frame used for given mode
- All surveys suffer from nonresponse but higher in self- than *Iwer*-administered modes
- Self-administration seems to increase honest responding for sensitive *Q*s; computerized self-administration confers extra advantage

Mode Comparisons

- “Mode effects” occur when same questions administered in two or more modes and results differ, despite holding constant all or most other features of design
  - Often more than mode varies between groups, so *confounding* is a problem
- Detecting mode effects requires random assignment of *R*s to modes
  - Even with random assignment, what appear to be measurement effects may be due to differential non-response or coverage
  - Collectively known as “selection effects”
- Many published mode comparison studies
  - Pairs of reasonable alternatives typically compared
  - Studies don’t exist on all combinations of all modes
  - Few with validation data so hard to know which mode produces more accurate estimates – just that the estimates are different

Some Mode Comparison Strategies

- In longitudinal panel study, assign *R*s to different modes in second wave
- Use different frames for each mode, e.g., area sample for in-person (FTF) and RDD sample for phone
- Use frame with information for >1 mode
  - e.g., University registrar may provide students' email address and phone number allowing researcher to randomly assign students in a single frame to either web or phone

In-Person vs. Telephone Interviews 1: Groves & Kahn (1979)

- National area probability sample vs. national RDD sample
  - Face-to-face questionnaire about twice as long as telephone questionnaire but same content
  - National field interviewing staff vs. centralized telephone staff
- Key Findings
  - Lower response rate on telephone (57%) than personal visit (74%)
  - More *R*s prefer FTF survey
  - Telephone interviews are faster paced, responses to open questions shorter
  - Few differences in substantive responses – suggests data quality similar in both modes

Personal vs. Telephone Interviews 2: Cannell et al. (1987)

- In-Person (FTF) administration of NHIS by Census Bureau compared to Michigan SRC Telephone Health Interview Survey
  - Order of questions different across modes
- Key Findings
  - Lower response rates on telephone (80%) than FTF (96%):
  - Higher reports of health events on telephone
  - Size of substantive differences between modes generally small
- This study, together with Groves & Kahn (1979), helped spur the growth of telephone surveys in U.S.

In-Person Interview vs. Self-administered Q'aire: Tourangeau & Smith (1996)

- Area probability sample in Cook County, Illinois
  - *R*s 18-45 years old
  - Questionnaire included items about sensitive topics
    - e.g., sexually transmitted diseases, marriage and cohabitation, pregnancy, sex partners, sexual attitudes, illicit drug use
  - *R*s randomly assigned to data collection mode: CAPI, CASI, or ACASI
- Key Finding
  - Self-administration uniformly results in more "socially undesirable" reports
  - Appears more undesirable responses in ACASI than CASI

Personal vs. Self-administered: Tourangeau & Smith (1996)

Ratios Comparing Levels of Reporting by Mode of Data Collection

	ACASI vs. CAPI	CASI vs. CAPI
Marijuana		
Lifetime	1.48	1.29
Past year	1.61	.99
Past month	1.66	1.19
Cocaine		
Lifetime	1.81	1.01
Past year	2.84	1.37
Past month	1.81	.95

In-Person vs. Telephone vs. Mail: deLeeuw (1992)

- Sample from telephone directory of 5 municipalities in Netherlands
- Compared quality of data collected in In-Person (FTF), Telephone, and Mail surveys
  - Same 26 interviewers randomly split between In-Person and Telephone, doing half of each before switching
- Findings
  - Longer responses to open-ended *Q*s in interview modes (In-Person and Telephone) than Mail, suggesting *R*s more conscientious in interviews
  - Increased reporting of loneliness and unhappiness in Mail, suggesting greater disclosure of negative information when *R*s self-administer *Q*s

In-Person vs. Telephone vs. Mail: deLeeuw (1992)

Missing Data and Precision of Reported Income by Mode

	Mail	Personal	Telephone
Missing data on income question (%)[1]	14	17	18
Mean reported in income[1]	2,953	2,628	2,758
Reporting income in: (%)[2]			
Guilders and cents	16	4	2
Rounded guilders	61	35	49
Approximate	23	61	49

[1]Mode differences not significant at $p>0.1$
[2]Mode differences significant at $p<0.01$

Telephone vs. Self-administered (IVR and Web): Kreuter, Presser, & Tourangeau (2009)

- Three modes: Phone, IVR, and Web
- Data from university records used to assess accuracy of responses
- Asked alumni (*R*s) sensitive questions about academic performance
  - e.g., “Did you ever receive a grade of D or F for a class?”
- Findings
  - Web: more reports of sensitive academic outcomes than Phone
  - Web: higher accuracy of reporting sensitive information than Phone, confirming assumption that (at least here) “more is better”
  - IVR in middle

Telephone vs. Web:
Pew Research (Keeter, 2015)

- Members of Pew's American Trends Panel randomly assigned to either Web or Telephone interview
- Differences in responses between modes common across the questionnaire but often not large: ~5% overall
- Among the mode effects:
  - Phone *R*s report greater (18-20%) satisfaction with quality of family and social life
  - Phone *R*s more likely to report that gays and lesbians, Hispanics and blacks face a lot of discrimination
  - Web *R*s more likely to rate various political figures as "very unfavorable"

Summary of Mode Comparison Studies

- Many published comparisons of how mode affects data quality
  - difficult to disentangle effects of mode on measurement from selection effects and different procedures between modes, e.g., question wording
- Robust effects:
  - Telephone interviews produce data quality at least as high as In-Person (FTF) interviews, but response rates are lower
  - Self-administration increases disclosure of sensitive info compared to *Iwer* administration
    - but may reduce effort invested by *R*s*
  - Web leads to levels of disclosure greater than other self-administered modes (e.g., IVR) but concerns about coverage and response rates*

*to be discussed in coming weeks